CANCER BIOLOGY

LECTURE
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OBJECTIVES

At the end of the lecture, student would be able to:

- Define Neoplasm and cancers.
- Describe the characteristic features of neoplastic cells.
- Explain the types of neoplasms.
- Understand the difference between Benign and Malignant tumor.
- Describe the nomenclature of tumor.
- List the causes of neoplasms.
- Know about the tumor markers.

Neoplasia

- **Neoplasm - new growth**
- **Tumor - Swelling**

Definition:

- **Abnormal mass of tissue**
- **Growth exceed and is uncoordinated with normal tissue**
- **Persists in the same excessive manner after cessation of initiating stimuli**
- **Grows without purpose**
- **Harmful to body**
- **Grows at the expense of normal tissue**
- **Molecular level – mass of abnormal cells with altered genome.**

NEOPLASM

- ***"A neoplasm is an abnormal mass of tissue, the growth of which exceeds and is uncoordinated with that of the normal tissues and persists in the same excessive manner after cessation of the stimuli which evoked the change"***

EPIDEMIOLOGY

Q: RANK OF CANCERS IN LEADING CAUSES OF DEATH?

Q: TOP THREE COMMMONEST CANCERS IN MALES & FEMALES?

Q: TOP THREE CANCERS CAUSING MORTALITY IN FEMALES AND MALES?

Characteristic of Neoplastic cells

a. Loss of responsiveness to normal growth controls .
b. Behave as parasites and compete with normal cells and tissues for their metabolic needs.
c. They are transformed i:e. they continue to grow regardless of normal regulatory controls.
d. They steadily increase in size regardless of their local environment and nutritional status of host.
e. They are dependant on host for their nutrition and blood supply.

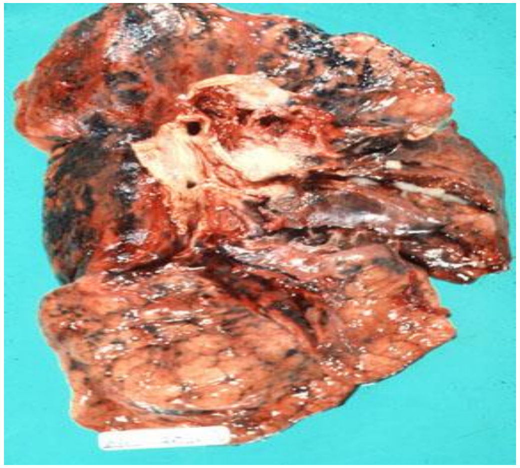
Types of Neoplasms

1. **Benign Tumors:** They remain localized, cannot spread to other sites and are amenable to local surgical removal and patient survives.

CANCER

The term “cancer” is usually associated with malignant tumors.

2. Malignant Tumors : They invade and destroy adjacent structures and spread to distant sites (metastasize) to cause death of patient.



Benign vs. Malignant

- **Slow growing**
- **Encapsulated**
- **Expansile growth**
- **No Metastasis**
- **Well Differentiated**

- **Rapidly growing (**display diminished growth control)
- **Non encapsulated**
- **Infiltrative growth**
- **Metastasis(**invade local tissues and spread, or metastasize, to other parts of the body)
- **Well-Poorly differentiated**
- **Stimulate local angiogenesis**

Key Properties

Cancer cells are characterized by :

(1) proliferate rapidly

(2) display diminished growth control

(3) display increased genomic mutations at the level of the nucleotide, small and large insertions and deletions, and gross chromosomal rearrangements, duplications and loss

(4) display loss of contact inhibition in culture in vitro

cont..

(5) invade local tissues and spread, or metastasize, to other parts of the body

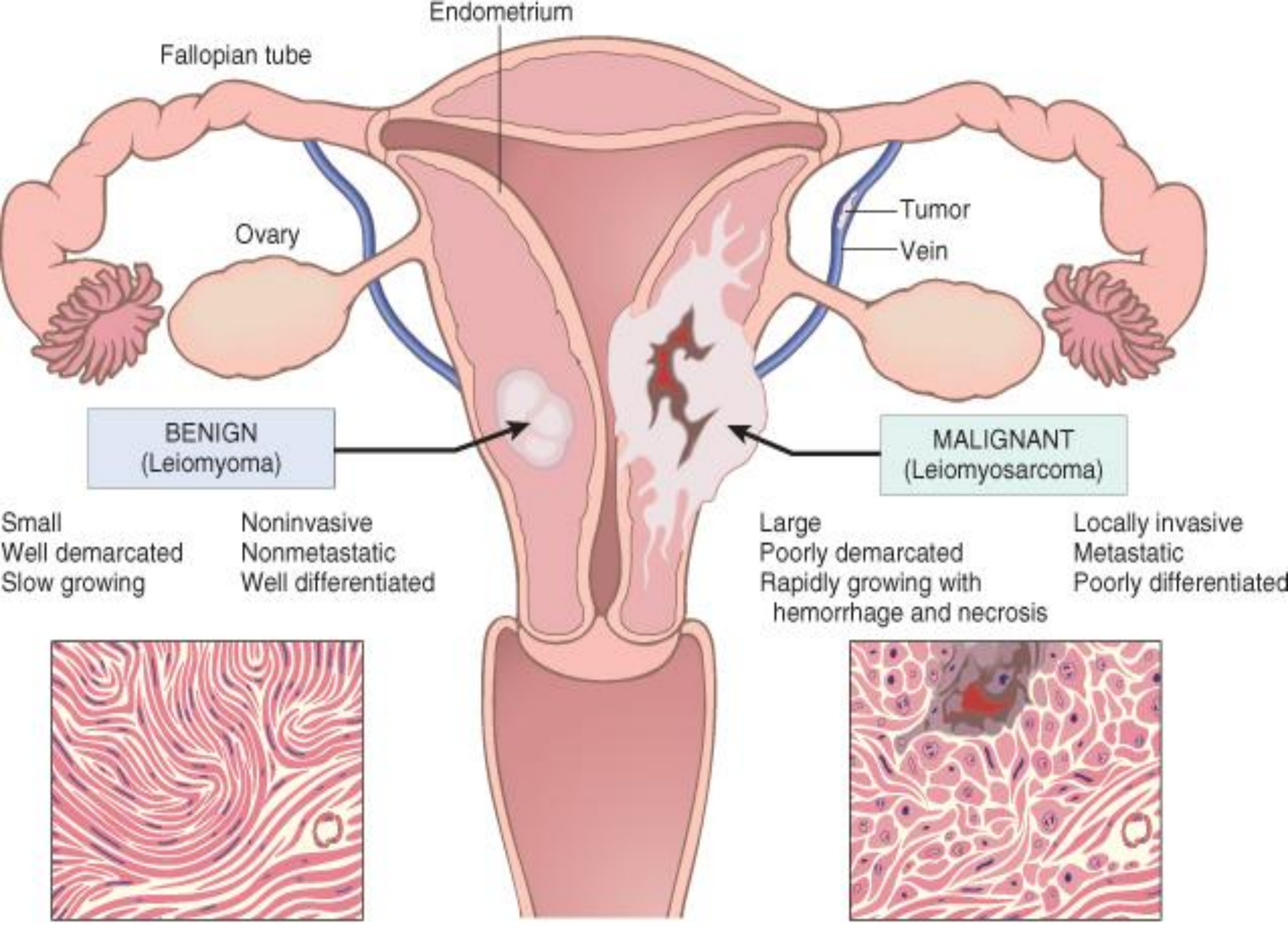
(6) are self-sufficient in growth signals and are insensitive to antigrowth signals

(7) stimulate local angiogenesis; and

(8) are often able to evade apoptosis.

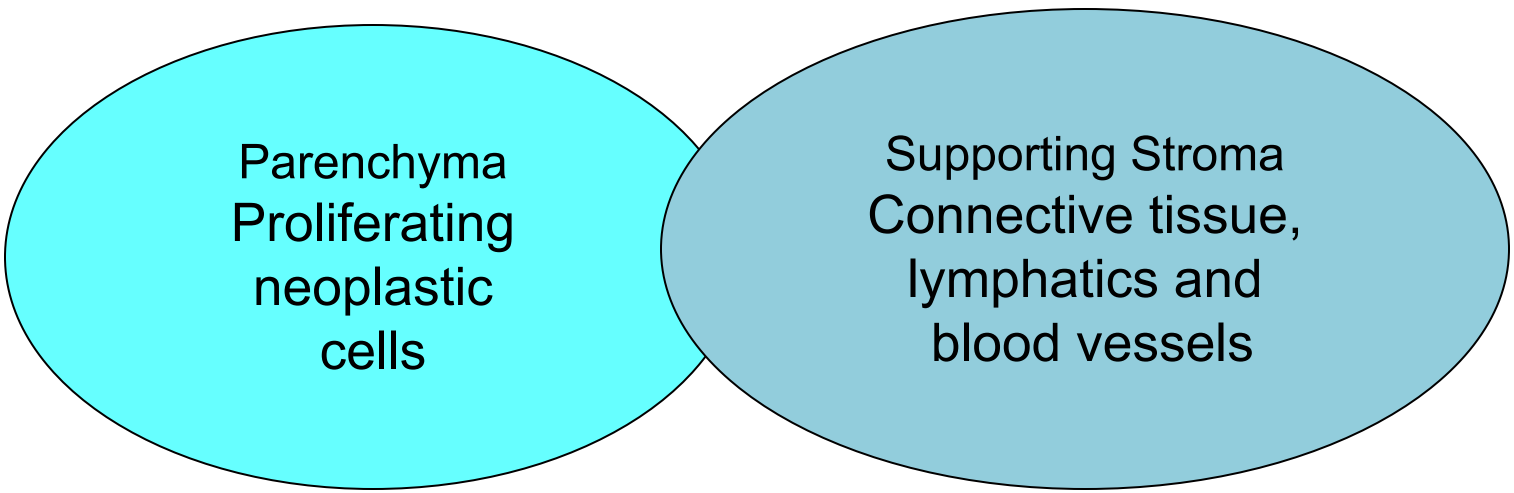
These properties are characteristic of cells of malignant tumors.

Benign lipoma on the serosal surface of the small intestine. It has the characteristics of a benign neoplasm: it is well circumscribed, slow growing, non-invasive, and closely resembles the tissue of origin (fat).


Endometrium
Fallopian tube
Ovary
Tumor
Vein
BENIGN
(Leiomyoma)
MALIGNANT
(Leiomyosarcoma)
Small
Well demarcated
Slow growing
Noninvasive
Nonmetastatic
Well differentiated
Large
Poorly demarcated
Rapidly growing with
hemorrhage and necrosis
Locally invasive
Metastatic
Poorly differentiated

Nomenclature Of Tumors

- **All tumors consist of two components:**



Nomenclature Of Tumors cont..

- Tumor nomenclature is based on behavior and parenchyma of tumor (cell of origin).

  Behavior - benign, not likely to kill
  malignant – likely to kill (cancers)

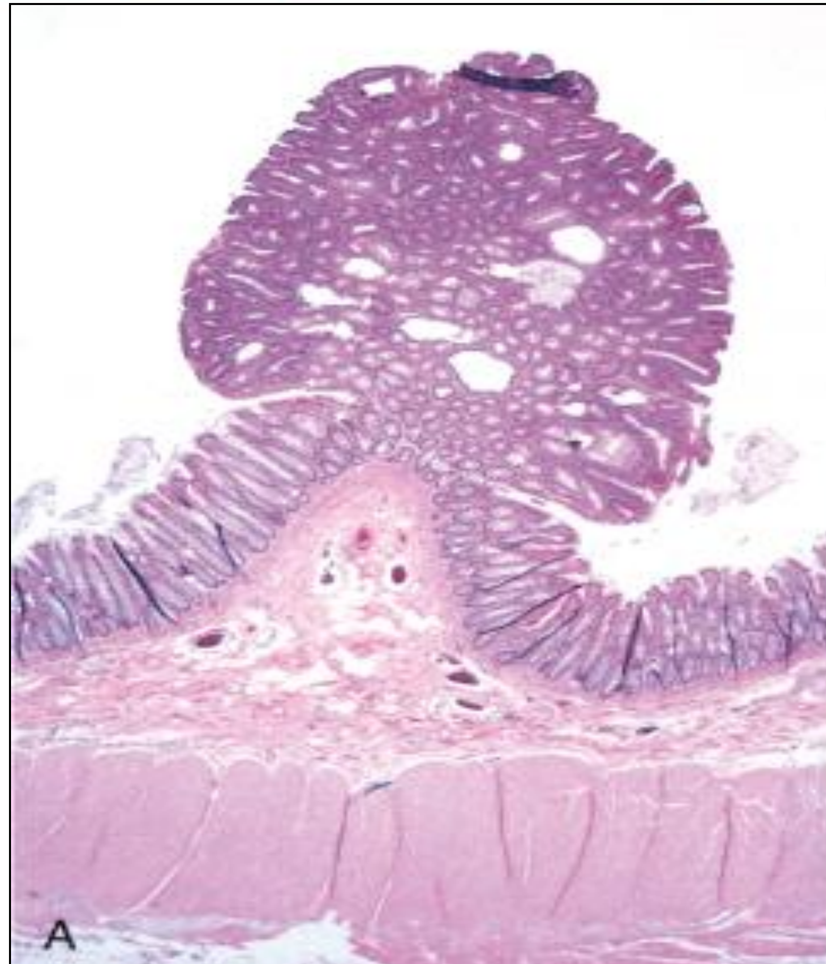
  Histogenesis- epithelia
  mesenchyme

Nomenclature of Benign Tumors

- **Suffix "oma" indicates a benign tumor**
- **Benign tumors of epithelial tissue e.g. adenoma arise from glands**
- **Benign tumors of mesenchymal tissues origin arise from mesoderm e.g. fibroma from fibrous tissue, Chondroma from cartilaginous tissue**

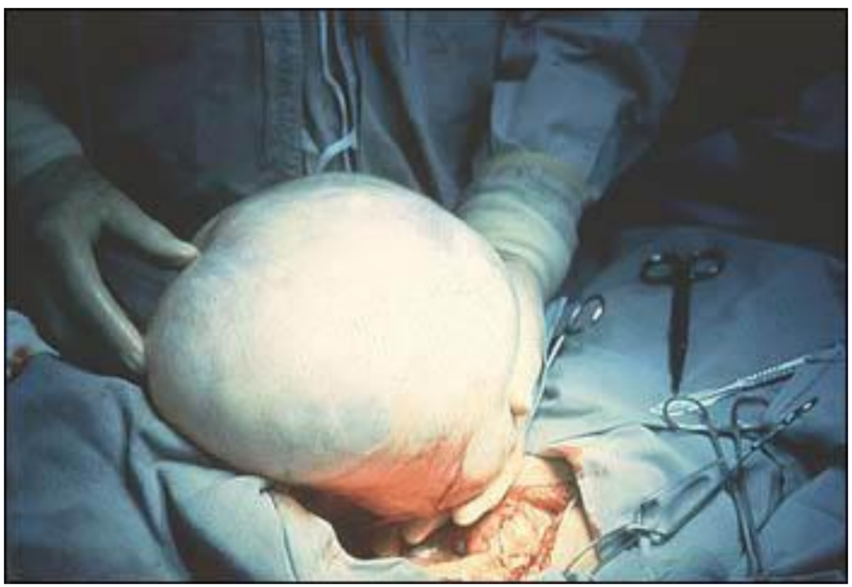
Papilloma finger like projections

Polyp



Cystadenoma

- Benign tumor that produces a hollow cystic mass.



Nomenclature of Malignant Tumors

- **Carcinoma- Malignant epithelial e:g. renal cell carcinoma, bronchogenic carcinoma, adenocarcinoma, squamous cell carcinoma and basal cell carcinoma.**

- **Sarcoma- Malignant mesenchymal, e:g. Fibrosarcoma, Chondrosarcoma.**

Tissue of origin	Benign	Malignant
Mesenchymal/ connective tissue	Fibroma Lipoma Chondroma Osteoma	Fibrosarcoma Liposarcoma Chondrosarcoma Osteogenic sarcoma
Endothelial and related tissues	Hemangioma Lymphangioma Meningioma	Angiosarcoma Lymphangiosarcoma Synovial sarcoma Mesothelioma Invasive meningioma
Hematopoietic		Leukemias Lymphomas

Tissue of origin	Benign	Malignant
Smooth Muscle	Leiomyoma	Leiomyosarcoma
Skeletal Muscla	Rhabdomyoma	Rhabdomyosarcoma
Epithelial	Squamous papilloma Adenoma Papilloma Cystadenoma Bronchial adenoma Renal tubular adenoma Liver cell adenoma Transititonal cell papilloma Hydatiform mole	SCC or epidermoid CA BCC Adenocarcinoma Papillary carcinoma Cystadenocarcinoma Bronchogenic carcinoma Renal cell carcinoma Hepatocellular carcinoma Transitional cell carcinoma Choriocarcinoma Seminoma Embryonal CA

More than one neoplastic cell- MIXED TUMORS		
Salivary gland	**Pleomorphic adenoma**	**Malignant mixed tumor of salivary gland origin**
Renal		**Wilms tumor**
Teratogenous (from more than one germ cell layer)		
Totipotential cells	**Mature teratoma/ dermoid cyst**	**Immature teratoma, teratocarcinoma**

Causes of Cancer

Environmental

- Chemical: aromatic hydrocarbons (smoked meat, cigarettes), azo dyes (bladder), aflatoxin (liver), asbestos (mesothelioma)
- Radiation: UVR, X-rays, rays produced by nuclear fission reactions.
- Viral: HPV (16,18, 31, 33, 35, 51), EBV (Burkitt), HBV, HTLV!

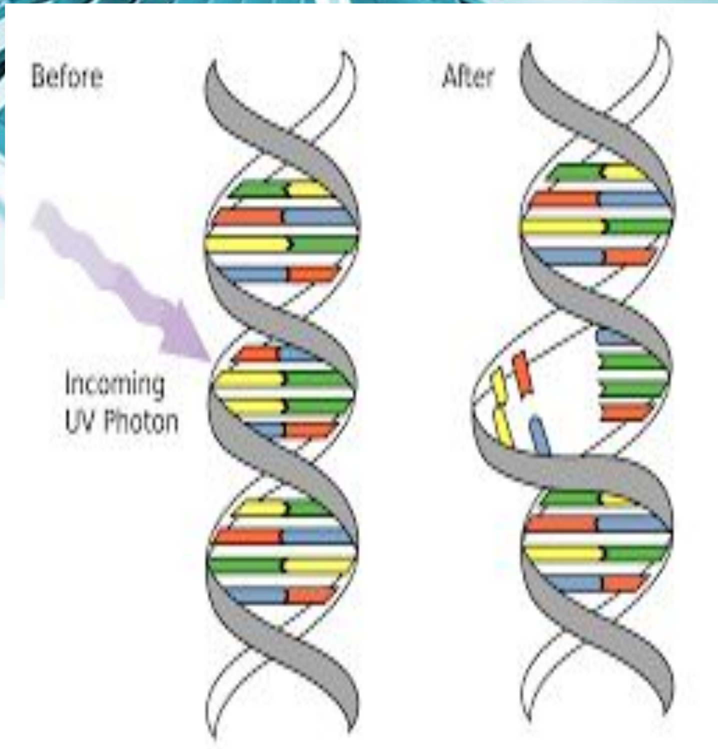
ROLE OF ONCOGENES AND TUMOR SUPPRESSOR GENES IN CANCER

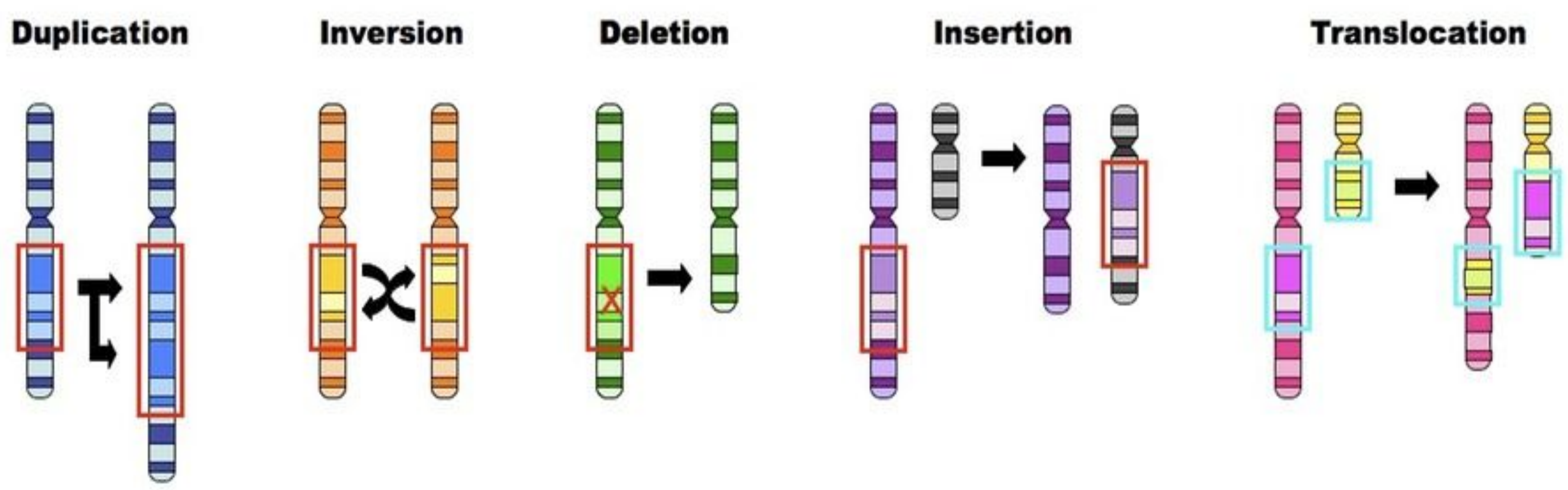
- Genetic changes involve mutations in genes normally involved in controlling cell division known as:
- **PROTOONCOGENES**
- **TUMOR SUPPRESSOR GENES (Anti-oncogenes)**

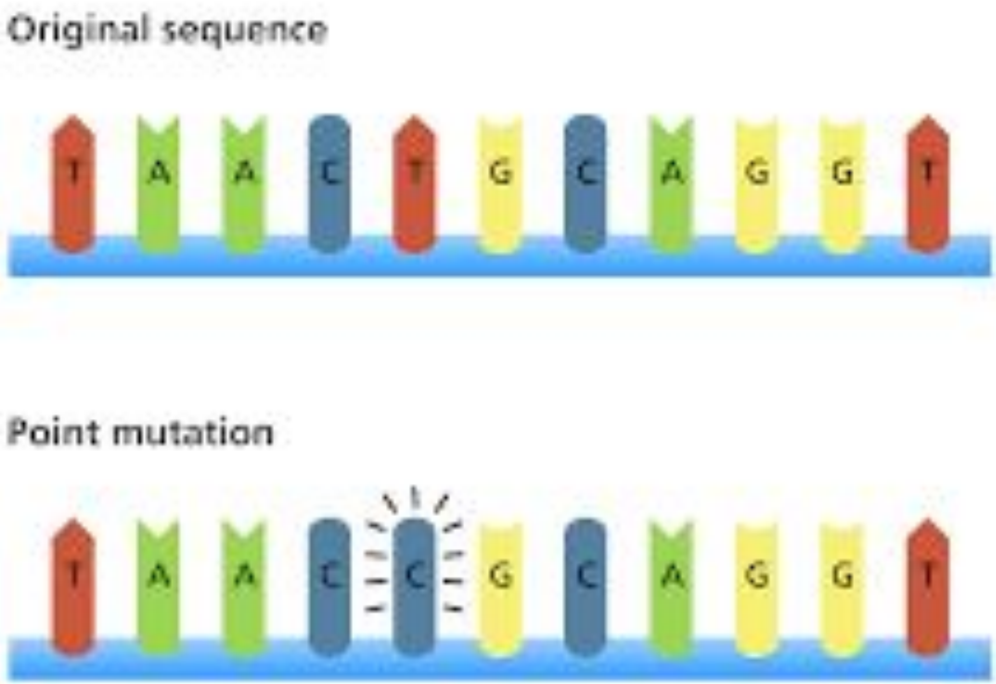
PROTO ONCOGENES

- Protooncogenes products include proteins that include:
  - **Growth hormones**
  - **Growth hormone receptors**
  - **Signal transduction molecules**
  - **Cell cycle progression proteins**
  - **DNA transcription factors**

- Changes in the quantity or quality of PO can occur due to:
  - **Point mutations**
  - **Translocations**
  - **Gene amplification**
  - **Deletions**



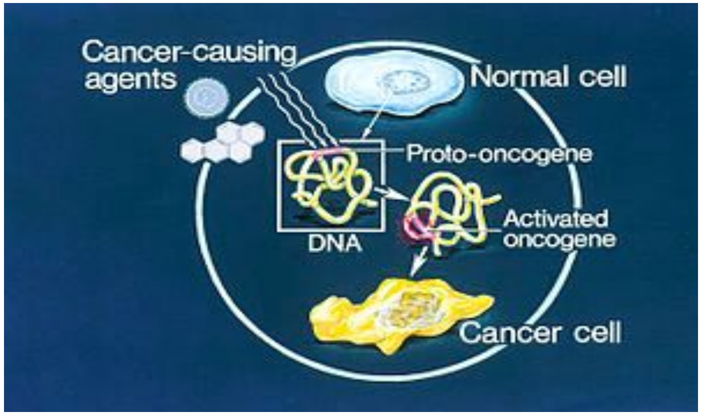



Original sequence
T A A C T G C A G G T
Point mutation
T A A C C G C A G G T

ONCOGENES

- Defective proto-oncogenes found in cancer cells are known as ONCOGENES.
  - **<u>ONCOGENES:</u>**
    - Def: “ An oncogene is a gene that has the potential to cause cancer”.

    In tumor cells, they are often mutated or expressed at high levels



TUMOR SUPPRESSOR GENES

- Def: “Is a gene that protects a cell from one step on the path to cancer”.
- Function: Repressive effect on the regulation of cell cycle or promote apoptosis.
- Loss of TSG, due to deletion or mutational inactivation contributes in the conversion of a normal cell to a transformed phenotype.

“Thus cancer results from multistep progression of a normal cell to a cancer cell that involves the gradual accumulation of mutations in protooncogenes and TSG”.

TUMOR MARKERS

Tumor markers are defined as

"Cellular characteristics or substances produced by cancer cells that can be used to differentiate tumor cells from normal tissue".

or

Produced directly by the tumor or as an effect of the tumor on healthy tissue.

- Concentration increases with tumor progression, highest levels when tumors metastasize

Tumor Marker Detection

Ideal TM include:

- ✔ Specificity for a single type of cancer (Tumor specific)
- ✔ High sensitivity for cancerous growth
- ✔ Correlation of marker level with tumor size
- ✔ Short half life of marker
- ✔ A substance that is released directly into the bloodstream detectable at small concentrations
- ✔ Absent in healthy individuals
- ✔ Readily detectable in body fluids.

- Unfortunately, all of the presently available tumor markers do not fit this ideal model.

TUMOR MARKERS

- Include diverse molecules such as
  - ENZYMES
  - HORMONES
  - FETAL ANTIGENS
  - CARCINOMA ASSOCIATED ANTIGENS
  - PROTEINS

USE:

- **Tumor markers are most useful for assisting in:**
  - Differential diagnosis
  - Staging of cancer
  - Estimating tumor volume
  - Monitoring disease progression
  - Detecting recurrence of cancer
  - Evaluating response to drug treatment regimens.

Tumor markers: **Enzymes**

- Increase due to metabolic demands of cells
- Indicate tumor burden
- Examples
  - Alkaline phosphatase
    - Bone, liver, intestine
  - Creatine kinase
    - Prostate, lung, breast, colon, ovarian
  - Lactate dehydrogenase
    - Liver, lymphomas, leukemias
  - Prostatic acid phosphatase
    - Prostate

FREQUENTLY ORDERED TUMOR MARKERS:
TISSUE SPECIFIC ANTIGEN

- **Prostate Specific Antigen (PSA)**
  - Produced in the epithelial cells of the prostatic ducts
  - In healthy men, some amounts of PSA can be detected
  - PSA is elevated in prostate infection, prostate cancer and benign prostate enlargement
  - PSA is usually higher in prostate cancer, useful for monitoring for reoccurrence of prostate cancer following prostatectomy.

Tumor markers:
Endocrine/ Hormones

- Eutopic hormones:
  - Beta-human chorionic gonadotropin in trophoblastic carcinoma and germ cell tumors.
  - Calcitonin by thyroid medullary tumors

- Ectopic hormones:
  - Adrenocorticotropic hormone (ACTH) by small cell carcinoma of lung, pancreatic, breast, gastric and colon cancer.

Severe systemic effects of hormones are called as Paraneoplastic syndromes.

Tumor markers:
Oncofetal antigens

- Considered normal in fetal development
- Become detectable in tumor formation
- Examples
  - Carcino-embryonic antigen(CEA)
  - Alpha-fetoprotein

Frequently Ordered Tumor Markers

- **Carcinoembryonic Antigen (CEA)**
  - Expressed during fetal development then re-expressed in tumor growth
  - Clinical use
    - Used for staging of colorectal, lung, breast ovarian, and GI cancers
    - Monitor therapy

Frequently Ordered Tumor Markers

- **Alpha(α) – Fetoprotein (AFP)**
  - Synthesized by the fetal liver
  - Re-expresses in certain types of tumors
  - Normally functions as a transport protein and helps to regulate oncotic pressure in the fetus

  - Most specific marker for hepatocellular carcinoma. Also found in germ cell tumors (testes, ovaries)

Tumor markers: **Proteins**

- Used to monitor therapy
- Examples
  - Beta-2-Microglobulin
  - Immunoglobulins: IgM with B cell cancer

Notable mentions

- **Breast cancer**
  - CA 15-3
    - Monitoring
  - CA 27.29
    - Monitoring
- **Ovarian cancer**
  - CA 125
    - Monitoring
- **Pancreatic cancer**
  - CA19-9
    - Monitoring

THANK YOU